ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **How do Organisms Reproduce?** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	Copying errors produce small variations, which explains the statement.
2	**(d)** variation, and those from the cutting are identical to the parent	Sexual gives variation; vegetative does not.
3	**(b)** undifferentiated cells to produce missing parts	Specialised dividing cells are essential.
4	**(d)** A is false but R is true.	A is false: most organisms cannot reproduce by regeneration; R is true.
5	**(b)** spores need moisture to germinate	Water is needed to activate spores.
6	**(d)** shortage of nutrients and oxygen	The placenta is the exchange site.
7	**(a)** Both A and R are true and R is the correct explanation of A.	The scrotum keeps the testes cooler.
8	**(d)** 8	Three divisions give $2^3 = 8$.
9	**(a)** Both A and R are true and R is the correct explanation of A.	With a single parent and no gamete fusion there is no mixing of DNA, so the offspring are almost identical clones.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; they are different modes of asexual reproduction, and R does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Regeneration (1); It involves growth of a whole body from a body part, not an outgrowth (1).	1+1
12	Unisexual flower (1); Pollen must be moved from the male flower to the female flower (1).	1+1
13	Pollen grain (1); Formed in the anther (1).	1+1
14	Number of divisions = 6 (1); $2^6 = 64$ Amoebae (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Divisions in 2 hours = 8 (1); Number = $2^8 = 256$ (1); Binary fission (asexual) (1).	1+1+1
16	Plants have meristematic cells for new growth (1); Humans have complex organ systems (1); Human body parts cannot make a whole body (1).	1+1+1
17	Embryo would not implant (1); Pregnancy would not be established (1); Embryo could not be nourished (1).	1+1+1
	OR: Ovaries release the egg and secrete oestrogen (1); Oestrogen controls thickening of the uterine lining (1); Without it the lining is not formed or shed (1).	1+1+1
18	Pollen tube must grow a long distance (1); It grows through the style tissue (1); It delivers male gametes to the ovule (1).	1+1+1
	OR: Many spores are produced quickly (1); Thick walls help survival in harsh conditions (1); Many spores fail to find a suitable place (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Stamen (anther produces pollen) (1). (ii) Pollination (1). (iii) Ovary (1). (iv) Self-pollination: pollen from the anther reaches the stigma of the same flower (or another flower of the same plant); cross-pollination: pollen reaches the stigma of a flower of a different plant (1); agent: wind, water, insects or animals (any one) (1).	1+1+1+2
20	Moist and warm: rapid growth of mould (1); Dry: little or no growth (1); Moist and cold: slow growth (1); Spores germinate only with moisture and warmth (1); Thick-walled spores wait for suitable conditions (1).	1+1+1+1+1
	OR: (a) Cotyledons feed the young seedling until it can photosynthesise (2); (b) Many are lost during transfer, so more pollen raises success (2); (c) Egg carries nutrients for the early embryo (1).	2+2+1

Mistakes to avoid:

- Writing that Plasmodium reproduces by binary fission; it undergoes **multiple** fission, whereas Amoeba shows binary fission.
- Using 'fertilisation' for pollination; pollination is transfer of pollen, fertilisation is fusion of male and female gametes.
- Saying that fertilisation takes place in the uterus; it occurs in the oviduct and the zygote is implanted in the uterus.
- Confusing regeneration with reproduction by fragmentation; regeneration uses specialised cells to rebuild a missing part, e.g. Planaria, Hydra.
- Stating that asexual reproduction produces absolutely identical offspring; tiny variations are produced by errors in DNA copying.
- Believing that the testes lie inside the abdomen; they are in the scrotum because sperm formation requires a lower temperature.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **How do Organisms Reproduce?** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 16	$2^4 = 16$.
2	**(a)** Plasmodium - budding	Plasmodium divides by multiple fission.
3	**(d)** A is false but R is true.	A is false as banana is seedless; R is true.
4	**(b)** self-pollination can occur within a flower	A bagged flower cannot receive foreign pollen.
5	**(a)** the survival value of variation	Variation helps the species.
6	**(a)** Both A and R are true and R is the correct explanation of A.	Both true and R explains A.
7	**(d)** It grows towards the ovule through the style	It develops on the stigma and goes to the ovule.
8	**(d)** nearly identical to the parent	Vegetative propagation gives clones.
9	**(c)** A is true but R is false.	Testes lie outside the abdomen in the scrotum, where the temperature is lower, which is needed for sperm formation; R is false.
10	**(d)** A is false but R is true.	Fertilisation takes place in the oviduct (fallopian tube), so A is false; implantation in the uterus is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Sperms cannot reach the urethra (1); So fertilisation of the egg cannot take place (1).	1+1
12	One seed (1); Only a fertilised ovule becomes a seed (1).	1+1
13	Vegetative parts carry copied DNA of one parent (1); Seeds form by fusion of gametes from two sources giving new gene combinations (1).	1+1
14	Nutrients are supplied through the placenta in ready form (1); Wastes also go back through the placenta (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Egg is large with stored food for the embryo (1); Sperm is small and has a tail for swimming (1); This shows division of labour between the gametes (1).	1+1+1
16	Each gamete carries half the DNA of the parent (1); Fertilisation combines these halves (1); So the zygote has features from both parents (1).	1+1+1
17	Hydra: budding by a single parent (1); Flowering plant: pollination and fertilisation (1); Plants use seeds that give variation (1).	1+1+1
	OR: Cuttings are genetically identical (1); If one is susceptible all are (1); Variation in seed-grown plants gives some resistant plants (1).	1+1+1
18	Fruits help in seed dispersal (1); Juicy fruits attract animals (1); Winged fruits are carried by wind (1).	1+1+1
	OR: Placenta has villi in contact with the mother's blood spaces (1); Exchange occurs across thin walls by diffusion (1); The two blood circulations remain separate (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Vegetative propagation (1). (ii) The eyes (buds) (1). (iii) Budding (1). (iv) Banana and rose do not produce viable seeds (or seeds are slow) (1); the new plants are genetically identical to the parent, flower and fruit earlier and retain good characters (1).	1+1+1+2
20	Vegetative parts already have mature tissue (1); Flowering and fruiting occurs earlier (1); Plants are clones with the same genes (1); A disease can affect all plants (1); Farmers can lose the entire crop (1).	1+1+1+1+1
	OR: Binary: division into two (Amoeba) (1); Multiple: many daughter cells (Plasmodium) (1); Budding: outgrowth (Hydra, yeast) (1); Fragmentation: filament breaks (Spirogyra) (1); All are asexual with one parent (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that Plasmodium reproduces by binary fission; it undergoes **multiple** fission, whereas Amoeba shows binary fission.
- Using 'fertilisation' for pollination; pollination is transfer of pollen, fertilisation is fusion of male and female gametes.
- Saying that fertilisation takes place in the uterus; it occurs in the oviduct and the zygote is implanted in the uterus.
- Confusing regeneration with reproduction by fragmentation; regeneration uses specialised cells to rebuild a missing part, e.g. Planaria, Hydra.
- Stating that asexual reproduction produces absolutely identical offspring; tiny variations are produced by errors in DNA copying.
- Believing that the testes lie inside the abdomen; they are in the scrotum because sperm formation requires a lower temperature.

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MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **How do Organisms Reproduce?** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Fertilisation, implantation, placenta formation, birth	Natural order.
2	**(d)** A is false but R is true.	A is false: Amoeba reproduces by binary fission; R is true.
3	**(d)** fusion of male and female gametes	Fertilisation.
4	**(a)** To attract pollinators	Insects help pollination.
5	**(a)** Both A and R are true and R is the correct explanation of A.	True with explanation.
6	**(c)** the middle of the cycle (around day 14)	Ovulation occurs mid-cycle.
7	**(b)** Producing gametes and sex hormones	Both are gonads.
8	**(a)** Both A and R are true and R is the correct explanation of A.	Tails give motility.
9	**(a)** Both A and R are true and R is the correct explanation of A.	R describes the function that makes the placenta essential.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true, but they describe different organisms; R does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Banana is a seedless variety (1); So it is grown vegetatively from suckers (1).	1+1
12	Self-pollination can occur (1); The flower is bisexual and compatible (1).	1+1
13	The uterine lining is retained for the embryo (1); It is not shed (1).	1+1
14	One parent cell copies its DNA (1); Variation arises only from small copying errors (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	In fragmentation, the body breaks and each piece grows into a new organism (1); In regeneration, a missing part is regrown by specialised cells (1); The two are related but the processes differ (1).	1+1+1
16	Sperm has 23 and egg has 23 chromosomes (1); Fertilisation fuses them (1); The zygote has 46 (1).	1+1+1
17	Bisexual: self-pollination is possible if pollinators are absent (1); Unisexual: forces cross-pollination and variation (1); Both forms help survival in different environments (1).	1+1+1
	OR: Place two slices, one open to air and one covered, in warm, moist conditions (1); Mould appears on the open slice (1); Spores from air germinate on moist bread (1).	1+1+1
18	They secrete fluids that make up semen (1); The fluid nourishes the sperms (1); It helps sperms to move (1).	1+1+1
	OR: Sperm small, egg large (1); Sperms produced in millions, eggs about one per month (1); Sperm is motile; egg is non-motile (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Oviduct (fallopian tube) (1). (ii) Placenta (1). (iii) Testosterone (1). (iv) Menstruation (1); the thick, spongy lining of the uterus prepared for the embryo is shed as the egg is not fertilised (1).	1+1+1+2
20	Testes produce sperms and testosterone (1); Vas deferens, prostate and seminal vesicles (1); Urethra and penis deliver semen (1); Body temperature is higher than sperm production needs (1); Scrotum keeps testes about 2 to 3 degree Celsius cooler (1).	1+1+1+1+1
	OR: Lining thickens after ovulation (1); Pregnant woman: embryo implants and lining is retained (1); Menstruation stops during pregnancy (1); Non-pregnant: lining breaks and is shed as menstrual flow (1); Cycle repeats (1).	1+1+1+1+1

Mistakes to avoid:

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- Saying that fertilisation takes place in the uterus; it occurs in the oviduct and the zygote is implanted in the uterus.
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